



Novel external-loop-airlift milliliter scale bioreactors for cell growth studies: Low cost design, CFD analysis and experimental characterization

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ARTICLE INFO

Keywords:

Scaling-up
Milliliter-scale
Airlift bioreactors
CFD
Optical sensors

ABSTRACT

Many researchers have limited access to fully equipped laboratory-scale batch bioreactors and chemostats due to their relatively high cost. This becomes particularly prohibitive when multiple replicas of the same experiment are required, but not enough bioreactors are available to operate simultaneously. Additionally, experiments using shaken flasks are common but show significant limitations in terms of maintaining homogeneous conditions in liquid cultures or installing instrumentation for monitoring. Here, we proposed to tackle this significant hurdle by providing a route to make available the manufacture of low-cost, milliliter-scale bioreactors. This approach seems plausible for enabling proof-of-concept experiments before moving to a larger scale without significant investments. The conceptually designed systems were based on external-loop bioreactors due to their flexibility, simplicity, and ease of assembling and testing. Designs were initially evaluated *in silico* with the aid of COMSOL Multiphysics. The successfully evaluated systems were then constructed via additive manufacturing and assembled for hydrodynamics testing via tracer methods. This was enabled by a newly home-made optical absorbance sensor (OAS) for in-line and real-time measurements. Both the *in silico* and experimental results indicated close to ideal mixing conditions and low shear stress. Cell growth curves were prepared by culturing *Escherichia coli* and following its cell density in real-time. Our cell growth rate and maximum cell density were similar to those previously obtained in closely related systems. Therefore, the proposed bioreactors are an affordable alternative for batch and continuous cell growth studies rapidly and inexpensively.

1. Introduction

The design, manufacturing, and testing of bioreactors with volumes in the order of milliliters is an approach that is becoming increasingly common for research laboratories. These smaller-scale systems allow saving resources associated with expensive reagents or facilitating the handling of lower quantities of toxic compounds (Allonneau et al., 2015). This is particularly valid during the early stages of research when uncertainty usually prevents significant capital investment in equipment (Brecht, 2009; Bramsiepe et al., 2012; Betts and Baganz, 2006). Along with the milliliter scale, microliter and liter sizes have also been considered as attractive alternatives for inexpensive prototyping of

bioprocesses (Betts and Baganz, 2006; Zhang et al., 2009; Schäpper et al., 2009). The selection depends on the offered comparative advantages for a specific application, in which case, one will be more suitable than the others. Once a bioprocess is effectively prototyped at a smaller scale, the analyzed key performance parameters can be effectively used to propose scaling-up schemes (Weuster-Botz et al., 2005; Neubauer et al., 2013; Marques et al., 2010). This, in turn, is critical to assure that processes can eventually reach the commercial scale.

Research on bioprocess engineering often occurs in milliliter-scale vessels for studying cell growth or production of metabolites. A common approach is to use flasks in a shaker incubator for different purposes, including studies of microbial consortia (Janbandhu and Fulekar,

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<https://doi.org/10.1016/j.jbiotec.2020.09.022>

Received 17 April 2020; Received in revised form 11 September 2020; Accepted 23 September 2020

Available online 28 September 2020

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2011; Kiliç et al., 2007), optimization of product biosynthesis (Khanna and Srivastava, 2005a,b; Bundale et al., 2015), novel media formulations (Salman et al., 2016; Khanna and Srivastava, 2005a,b), studies of metabolism and metabolic response (Olson et al., 2016; Gasser et al., 2007; Lu et al., 2010), and analysis of inhibitors (Reyes et al., 2011; Almario et al., 2013; Xiong et al., 2018). A drawback of this approach is that instrumenting these setups is relatively complicated due to the absence of appropriate sensors and control systems (Losen et al., 2004). This issue is particularly critical in the case of bioreactor operation parameters such as dissolved oxygen and pH, which are fundamental for the productivity of the process (Ryan et al., 1989; Willow and Cohen, 2003; Painter and Loveless, 1983). As a result, bioprocesses often suffer from low reproducibility when changing from shaken to stirred bioreactors (Nunes et al., 2013). This has been attributed to the differences in mixing patterns and mass transfer coefficients from one system to the another (Vrabel et al., 2000; Ma et al., 2005). In this regard, shaken flasks generally offer poor mixing and limited oxygen transfer (Klöckner and Büchs, 2012). These differences detrimentally impact the scaling-up schemes as optimal conditions are not necessarily reproducible at larger scales or in bioreactors with better geometrical configurations and control (Kumar et al., 2004a,b). As a better approach, automatically controlled bench-scale bioreactors (about 5 L in volume) have been widely implemented as a more robust strategy to predict the behavior at a larger scale (Riedlberger and Weuster-Botz, 2012a,b; Zawada et al., 2011). These systems are equipped with suitable mixing systems, proper geometrical configurations, and monitoring probes, which generally results in a significant investment that is prohibitive for labs operating under low budgets. Another possible drawback of these systems is related to mixing. Under very specific conditions, such as high viscosity media because of the presence of extracellular polymers, achieving ideal mixing could be difficult even for bioreactors as small as 2 L (Gill et al., 2008). In general, larger bioreactors pose difficulty in maintaining the homogeneity of the medium, their operation is more expensive, and cleaning and sterilization sometimes are tedious and costly processes (Patnaik, 2015).

As an alternative, over the past two decades, a large body of research has been dedicated to exploring microliter size bioreactors, termed microbioreactors (Zanzotto et al., 2004; Schäpper et al., 2009; Maharbiz et al., 2004; Szita et al., 2005; Kostov et al., 2001). These devices offer the possibility of conducting biotransformations in ultra-low volume compartments, which results in attractive when the reagents or bioactive compounds are scarce, or the reactions proceed in a difficult-to-control manner. Despite these potential benefits, micro-bioreactors operate under a laminar flow regime, which differs from the turbulent regime usually maintained in larger scale bioreactors (Rohe et al., 2012; Lee et al., 2006; Schäpper et al., 2009). This derives in operation conditions that significantly vary from those used on larger scales. Additionally, the precision required for manufacturing these devices might involve the use of sophisticated and expensive cleanroom fabrication techniques (Schäpper et al., 2009). Moreover, proper instrumentation for these systems is often unavailable, challenging to implement, or unaffordable (Allonneau et al., 2015).

A scale between micro and bench is obtained when operating at the milliliter range (i.e., tens to hundreds of milliliters). On such a level, there is still a low consumption of reagents, but as opposed to devices at the microscale, the geometries can be made closer to those of the larger-scale bioreactors (Kumar et al., 2004a,b). This is beneficial from the operation viewpoint as most electronic components required for instrumentation are still available and affordable at such scale (Vallejos et al., 2006; De León et al., 1998). Moreover, if manufacturing is kept at a low cost, the assembly of multiple simultaneous reaction systems is a tangible possibility. This has been enabled by the emerging additive manufacturing techniques, like 3D printing, which have considerably declined in cost over the past few years (Peris et al., 2017; Xia and Fang, 2009). As a result, these platforms facilitate a wide variety of studies where recreating geometrical features is of the utmost importance. This

includes studies of productivity and optimization (Kheradmandnia et al., 2014; Seo et al., 2012), mass transfer evaluation (Lamping et al., 2003), nutrient consumption and cell growth (Villena and Gutiérrez-Correa, 2011), rheology (Zou et al., 2008), and training of personnel for bioreactor operations (Hass, 2016). Despite the potential advantages of these systems, achieving ideal mixing still requires a combination of parameters such as the geometrical configuration and the type of stirring and sparging systems (Li et al., 2008).

Respect to the geometrical configuration, perhaps one of the most straightforward and attractive alternatives for scaling-down is the stirred tank. This type of bioreactor offers superior mixing patterns, but it is challenging to monitor in a real-time manner with the aid of in-line sensors such as those based on optical density measurements. This is mainly due to the inhomogeneity of the media as it contains fine air bubbles dispersed throughout the volume of the reactor, which ultimately alters the transparency required for the accurate measurements (Fan et al., 2016; Zitzmann et al., 2018). Alternatively, air-agitated systems such as bubble and airlift reactors, equipped with external circulation loops, provide the opportunity for monitoring as sensors can be coupled directly to such loops (Šimčík et al., 2011; Utiger et al., 1999). In this case, the main challenge is to assure that bubbles passing through the loop are reduced to the minimum. This requires trial and error experiments where the type of sparging system and the airflow are adjusted. An avenue to accelerate this process is to evaluate the different operating conditions *in silico*. A reliable methodology for such analysis is the use of computational fluid dynamics (CFD) approaches.

With increasing computational power worldwide, CFD has become a widely used tool for developing and evaluating new bioreactor configurations (Hekmat et al., 2010; Morchain et al., 2013; Šimčík et al., 2011). Under this approach, the equations of conservation of momentum are numerically solved in a computational domain (resembling the bioreactor) that has been discretized with the aid of finite volumes or finite elements (Wesseling, 2009). This allows predicting mixing patterns and tracking particle trajectories inside the bioreactor (Hekmat et al., 2010; Morchain et al., 2013). Importantly, CFD has been successfully applied to modeling air-agitated systems such as bubble and airlift bioreactors. Several reports describe CFD as a fundamental tool for predicting the possible outcomes of varying the geometrical configurations and operating regimes on bubble distribution and mass transfer, circulation times, dead zones, mixing patterns, pressure changes, and shear stress (Allonneau et al., 2015; Hekmat et al., 2010). In the particular case of milliliter-sized bioreactors, the prevailing ideal mixing conditions and relatively small size might be useful to save computational time without losing accuracy (Morchain et al., 2013).

These ideal mixing conditions also allow the implementation of some models that apply to “real” bioreactors such as the N-continuous stirred tank reactors (CSTR) in series approach (Levenspiel, 1919). This method is advantageous because the experimental parameters obtained under ideal mixing conditions can be directly fitted to the “real” non-ideally mixed bioreactor model. As a result, the models’ validity needs to be assured experimentally by achieving ideal mixing conditions, which is more easily attainable at the milliliter scale. Due to its simplicity, the tracer test method has been widely used for assessing the mixing regime and residence time distributions of bioreactors (Levenspiel, 2012). In this approach, a tracer substance is injected into the system and tracked in time as it circulates. Some of the most reliable tracer molecules include strong acids or bases, redox pairs, dyes and fluorophores, and salts (Smits and Lim, 2012). The technique focuses on establishing the hydraulic retention time (HRT) distribution, and the subsequent fitting it to the N-continuous CSTR in series model (Betts and Baganz, 2006). Thus far, however, most hydrodynamics-related studies have been performed on liter-scale bioreactors, while studies for milliliter-scale bioreactors are rather scarce (Allonneau et al., 2015; Riedlberger et al., 2012; Riedlberger and Weuster-Botz, 2012a,b).

Here, we aimed at evaluating the hydrodynamics of our recently conceived milliliter-scale, external-loop, airlift bioreactor. This system

was designed to maintain ideal mixing conditions while coupling biomass monitoring devices to the system, which is a difficult task in the conventional CSTR bioreactors. *In silico* characterization was accomplished by conducting multiphase CFD simulations in COMSOL Multiphysics. Hydraulic characterization was accomplished by implementing a tracer test method based on the injection of the dye bromocresol green. CFD results allowed us to identify an appropriate geometrical configuration, size, and airflow operation regime to assure that the number of bubbles returning through the downcomer is minimized. The obtained design served for assembling a prototype with parts fabricated with the aid of additive manufacturing techniques (Krujatz et al., 2016). This was crucial to confirm that it was feasible to install a custom-made colorimeter to monitor the optical density of cell culture in real-time. The tracer method results provided the operating conditions under which close to ideal mixing conditions could be achieved for the bioreactor. Finally, growth tests using *Escherichia coli* as a model organism were performed and successfully compared to previously reported kinetics data.

2. Materials and methods

2.1. Modeling and simulation

The fluid dynamics was studied *in silico* with the aid of the software COMSOL Multiphysics® (v.5.3.1). The simulations were run in a 12 Gb RAM, eight cores CPU, Lenovo Thinkpad computer. The purpose of these simulations was to assess if the proposed geometry would permit operation conditions such that a close to ideal mixing was possible but, at the same time, to estimate whether the relatively high air flows required might lead to excess bubbles passing through the downcomer, in which case the optical density readings could be biased.

Importantly, the main goal was to develop a bioreactor that maintains homogeneous conditions for the liquid culture while facilitating the installation of instrumentation. One important challenge was the in-

line measurement of the optical density, which was addressed by implementing the external loop geometry.

2.1.1. Computational domain

Fig. 1A shows the 2D computational domain implemented in COMSOL, which corresponds to a cross-sectional cut of the milliliter scale bioreactor. The geometry takes into account all the accessories required for assembling the system. Axes scales are in millimeters.

2.1.2. Governing equations, boundary conditions, material properties, and initial values

Governing equations for two-phase flow were implemented with the aid of the Turbulent Bubbly Flow module of COMSOL Multiphysics®. The Bubbly Flow Interface solves for the average volume fraction occupied by each phase instead of tracking each bubble precisely and independently. The main advantages of this model are its relatively low computational demand and the ease of running parametric studies, which helps to evaluate different configurations and operating conditions. According to this approach, the momentum transfer equation takes into account a $\kappa - \epsilon$ model for turbulence for the continuous phase, i.e., the liquid phase (Eq. (1)).

$$\phi_l \rho_l \frac{\partial u_l}{\partial t} + \phi_l \rho_l u_l \cdot \nabla u_l = -\nabla p + \nabla \cdot \left[\phi_l (\mu_l + \mu_t) \left(\nabla u_l + (\nabla u_l)^T - \frac{2}{3} (\nabla \cdot u_l) I \right) \right] + \phi_l \rho_l g + F \quad (1)$$

where ϕ_l is liquid phase volume fraction (m^3/m^3), ρ_l is the density of the liquid phase (kg/m^3), u_l is the velocity vector for the liquid phase (m/s), p is the pressure (Pa), g is the gravity vector (m^2/s^2), and F is any additional volume force acting on the system (N/m^3). Turbulence is modeled by incorporating the turbulent dynamic viscosity (μ_t), and the turbulent kinetic energy (κ) into the conservation equation. This allows

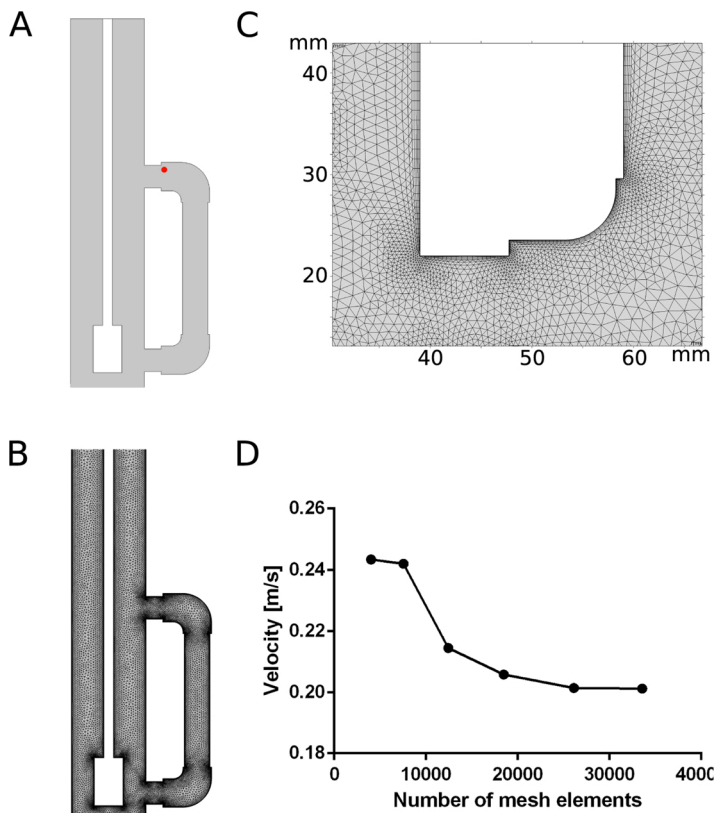


Fig. 1. Geometry implemented with respective mesh. A. Built computational domain in COMSOL Multiphysics®. The red dot indicates a point within the domain selected for mesh convergence analysis. B. Meshed computational domain with a final number of 34,000 elements. The minimum element size was 73 μm . C. Zoom in the bottom section of the computational domain. Dimensions of the computational domain correspond to those of the actual bioreactor. D. Mesh convergence analysis. The evaluated parameter was the average velocity at a selected point within the computational domain. No changes beyond 2 % in such parameter were observed above about 34,000 elements. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

accounting for effects such as convection and diffusion of turbulent energy.

The turbulent dynamic viscosity was modeled with the aid of Eq. (2), where C_μ is a model constant, and ε is the turbulent's energy dissipation rate. The transport for the turbulent kinetic energy (κ) is given by Eq. (3).

$$\mu_T = \rho_l C_\mu \frac{\kappa^2}{\varepsilon} \quad (2)$$

$$\rho_l \frac{\partial \kappa}{\partial t} + \rho_l (\vec{u}_l \cdot \nabla) \kappa = \nabla \cdot \left[\left(\mu + \frac{\mu_T}{\sigma_\kappa} \right) \nabla \kappa \right] + P_\kappa - \rho_l \varepsilon + S_\kappa \quad (3)$$

The evolution of the turbulent's energy dissipation rate is governed by Eq. (4).

$$\rho_l \frac{\partial \varepsilon}{\partial t} + \rho_l (\vec{u}_l \cdot \nabla) \varepsilon = \nabla \cdot \left[\left(\mu + \frac{\mu_T}{\sigma_\varepsilon} \right) \nabla \varepsilon \right] + C_{\varepsilon 1} \frac{\varepsilon}{k} - C_{\varepsilon 2} \rho_l \frac{\varepsilon^2}{k} + C_{\varepsilon} S_\varepsilon \quad (4)$$

where C_ε are the coefficients of destruction: one in steady-state and the other in motion. Both are empirical parameters. S_κ accounts for the bubble induced turbulence (Eq. (5)) and P_κ is the generation of turbulent kinetic energy; thus, it uses turbulent viscosity (Eq. (6)).

$$S_\kappa = -C_\kappa \phi_g \nabla p \vec{u}_{slip} \quad (5)$$

$$P_\kappa = \mu_T [\nabla \vec{u}_l : (\nabla \vec{u}_l + (\nabla \vec{u}_l)^T)] \quad (6)$$

Gravity volume force direction was considered in the negative Y-axis. Water density was taken as 998.2 (kg/m³) for operating conditions of 298 K and 1 atm.

The continuity equation for the Bubbly Flow interface is given by Eq. (7).

$$\frac{\partial}{\partial x} + (\rho_l \phi_l + \rho_g \phi_g) + \nabla \cdot (\rho_l \phi_l \vec{u}_l + \rho_g \phi_g \vec{u}_g) = 0 \quad (7)$$

where ϕ_g is gas phase volume fraction in (m³/m³), ρ_g is the density of the gas phase (kg/m³), and $\vec{u}_g$ is the velocity vector for the gas phase (m/s).

The transport equation for the volume fraction of gas is given by Eq. (8).

$$\frac{\partial y}{\partial x} + \nabla \cdot (\rho_g \phi_g \vec{u}_g) = -m_{gl} \quad (8)$$

where m_{gl} is the mass transfer rate from the gas to the liquid kg/(m³ s). The gas inlet boundary condition was set to different values for the hydrodynamics studies. These included 0.8, 1.0, 1.2, and 1.4 L/min. The outlet boundary was set to atmospheric pressure to assure a pressure drop that was enough for the bubbles to ascend from the diffuser to the system exhaust. The non-slip boundary condition was defined for the other boundaries.

To investigate whether the bubbles generated by the sparging system passed by the downcomer at levels such that they maintained the transparency needed for the in-line sensing of the reaction medium, the Particle Tracing module of COMSOL Multiphysics® was implemented to track the bubbles as they circulate in the bioreactor. This module provides a Lagrangian description for particle trajectories in the time domain. Newton's law of motion is solved for each component of the position vector in the time domain (Eq. (9)). As a result, for a 2D system, a set of two ordinary differential equations are solved for each particle. The particle-particle interactions were not taken into consideration to unambiguously determine particle positions. Bubble diameter was assumed to be 500 μ m with the properties of air at 298 K and 1 atm. Additionally, it was assumed that the diffuser produced all bubbles and that their size distribution was utterly uniform.

$$\frac{\partial}{\partial x} \left(m_p \frac{dq}{dt} \right) = F \quad (9)$$

where q is the particle position (m), m_p is the particle mass (kg), and F is the total force on the particles (N).

2.1.3. Mesh convergence analysis and solver

Mesh convergence analyses were conducted over six free triangular meshes ranging from 4000 to 34,000 elements. The evaluated variable was the velocity magnitude in one location in the downcomer (see the red dot in Fig. 1A). The convergence criterion was that upon increasing the number of elements in the computational domain, the velocity magnitude in the selected location remained within 2 % of the previous value. Fig. 1D shows that the convergence criterion was achieved for 34,000 elements. With such a level of discretization, the corresponding meshed domain is shown in Fig. 1B and C.

The simulation was conducted under transient conditions, and therefore, a time-dependent study was set up using a segregated approach for the turbulence parameters and a PARDISO direct solver for interfacial parameters, pressure, velocity, and effective density. This solver provides a high performance, robust, and memory-efficient direct method for solving large sparse symmetric and unsymmetrical systems of equations (Petra et al., 2014a, b).

The CFD solution was coupled to the particle tracing module, and in that case, the study was solved with the aid of iterative GMRES solver. The selected mesh was adequate to effectively track the bubbles circulating within the system because the elements had sizes below those of the bubbles.

2.2. Bioreactor manufacture

The main body of the bioreactor was made of polyester resin, a material that exhibits flexibility to incorporate geometrical changes as needed quickly. Additionally, this material is thermally resistant and can tolerate conventional autoclaving processes. The milliliter-scale bioreactor was conceptually designed as an external-loop-airlift, which was equipped with a riser column with a diameter of 39 mm and a height of 200 mm. The downcomer was a glass tube with a diameter of 14 mm and a height of 95 mm. The working volume was 170 mL (140 mm of liquid column), and the headspace was 72 mL (height = 60 mm). Air was supplied through a 5 mm stainless steel tube entering the bioreactor through a hole at the center of the top cap. A leak-proof silicone rubber gasket was attached to the inlet to avoid leakage. This tube was connected to a 13 mm $\times$ 20 mm aeration stone (as a diffuser) with a pore size of 500 μ m, located 10 mm above the bottom of the bioreactor. For operation, the airflow was set to 0.8 L/min (18 °C and 74.6 kPa). The temperature was maintained at 37 °C with the aid of an *ad hoc*-made incubator box equipped with an XH-W1209 temperature control module. This module was connected to a temperature-sensing probe (NTC 10 K 0.5 %), and a C1P PVC insulated heating cord (Dpstar group). Fig. 2 shows a schematic of the components of the milliliter-scale bioreactor.

The manufacture of the bioreactor proceeded with the aid of additive manufacturing techniques. Although various construction alternatives could be implemented, in this case, we proposed the use of polyester resin. Once the molds are available, the bioreactor's main structure can be replicated as many times as needed without the need for advanced technical skills or incurring in high investments. The polyester resin was selected for manufacture due to its thermal resistance (appealing for autoclavable parts) and the possibility to be cast on top of itself to create multilayer constructs by manual additive manufacturing. The obtained set of molds can be used as a robust platform for assembling different types of bioreactors, even if they require the development of parts with complex geometries.

The process started with the construction of hard models with the exact geometry to be replicated in resin. The main structure consisted of a glass vessel provided with a 3D printed threaded neck and a 3D printed threaded cap with ports. The external loop consisted of two 3D printed elbow connectors, silicon rubber gaskets, and a glass tube. Those parts

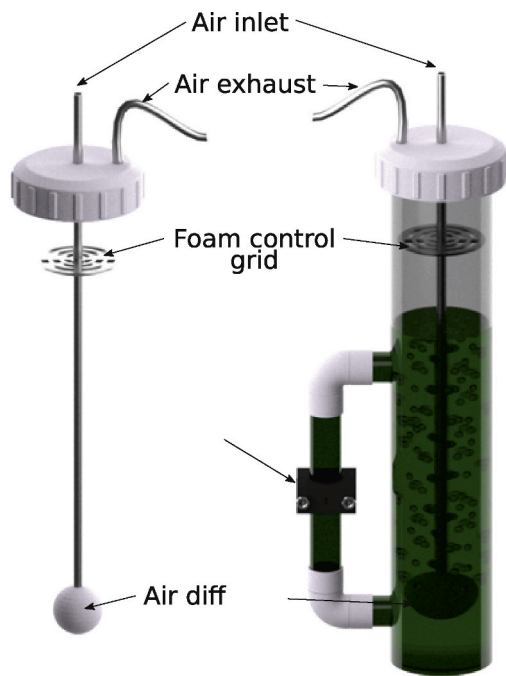


Fig. 2. Schematic of the milliliter-scale external-loop-airlift bioreactor. The bioreactor was mainly made of polyester resin except for the downcomer, where a glass tube was used to couple an optical sensor around it. The height of the air sparger was adjustable, and a grid for foam control was located at the top of the system on the air tube inlet. The total working volume of the bioreactor was 170 mL.

were printed using a Duplicator 4 printer (Wanhao, China) in polylactic acid (PLA) (3Dbots, USA). 3D printing was necessary due to the difficulty of finding standard size components for every single part of the hard models, so ad-hoc parts were necessary. Three different silicon rubber molds were made using the hard models: one for the main structure, one for the cap, and one for the elbow connectors. Since the resin is not transparent enough to permit appropriate optical density measurements in the downcomer, the glass tube was not used for making molds but directly incorporated into the final bioreactor assembly. Thanks to using the resin, the parts are cast individually, assembled, and bonded into a single structure with the aid of extra resin. This additive construction approach based on the resin facilitates replicating completely sealed vessels with complex geometries.

The HRT distribution by a tracer test was conducted in a “continuous version” of the bioreactor, which was assembled with a feeding tube inserted on the bioreactor cap. This tube had an adjustable height such that feeding was possible either by the top or the bottom of the raiser. Also, two exit tubes were installed at the top and the bottom of the raiser.

2.3. Optical absorbance sensor

One of the main advantages of the built external-loop-airlift was the possibility to monitor cell density in an in-line and real-time manner. This was accomplished by mounting an optical absorbance sensor (OAS) on a section of the transparent downcomer. This OAS (Fig. 2) comprises a detachable split case where one half holds a LED light source for emission at a fixed wavelength (630 nm, in this case). The second half contains a photoresistor, which is responsible for detecting the non-absorbed/non-deflected light. These two halves are screwed to each other around the glass tube. Incident light passes through the cell suspension flowing inside the downcomer. As cell density increases, light detected by the photoresistor decreases, which was then translated into a quantitative estimate of the cell density (see below for calibration

details). The detachable case was 3D-printed. The printing material was black PLA (3Dbots, USA) to reduce external light interference, further avoided by placing the system inside the incubator box. The sensor case was dismantled after each experiment to proceed with the autoclaving process. This procedure was to assure no damage to the case material as it failed to tolerate temperatures above 80 °C.

A circuit was developed *in-house* for automated data collection and storage in a solid-state memory card. Arbitrary units were assigned to the absorbance measurements collected with the sensor. A value of 999 corresponded to complete light saturation, while a value of 0 compared to complete darkness (total light absorption). The OAS was also calibrated against a Thermo Genesys 20 spectrophotometer (Thermo, USA) by measuring the absorbance of five bromocresol green (BG) in water solutions at five different concentrations. The obtained data was then correlated with the measurements via OAS according to the previously defined arbitrary units.

2.4. Tracer test

A stock solution (0.1 % (w/v) in ethanol) of BG was used as a tracer (Jimenez et al., 1988). An adapted version of the OAS was used for detecting the tracer at the bioreactor’s outlet. Accordingly, the OAS case was redimensioned to fit a glass tube of 7 mm of diameter connected to either the top or the bottom of the riser column. A peristaltic pump, *ad-hoc* designed and constructed in our laboratory, was used to feed water to the bioreactor through a stainless-steel tubing connected to the top cap at a flow rate of 14.6 mL/h. In this case, water served as reference fluid for proof-of-concept experiments which permitted to preliminarily evaluate if “ideal mixing” was achievable when using the bioreactor geometry previously tested *in silico*. At $t = 0$, a 1 mL pulse of the tracer stock solution was injected into the riser column using a 5 mL syringe. This was followed by instantly starting a continuous water feed. Colored water coming out of the bioreactor was analyzed with the aid of the OAS. The absorbance data was automatically collected every minute for 24 h. Four different setups were considered with alternating combinations of the feeding and the outlet connected to the top or the bottom of the raiser. Experiments were done in triplicate with three bioreactors operating simultaneously.

Data were converted to absorbance and subsequently to BG concentration with the aid of the calibration curve described above. Concentration profiles at the outlet of the bioreactors were analyzed according to Levenspiel (Levenspiel, 2019) by calculating the $E(t)$ probability distribution (Eq. (10)) and the hydraulic retention time (t_r) as the expected value (first momentum) of $E(t)$ (Eq. (11)). With the definition of $\theta = t/t_r$, $E(t)$ can be converted to $E(\theta)$ using Eq. (12). Finally, experimental $E(\theta)$ data were fitted to the N tanks-in-series model (Eq. (13)) to calculate the number of equivalent “ideally-mixed tanks” (N) that represent the hydrodynamics of the real bioreactors. In this case, all of the equivalent tanks have the same volume and retention time.

$$E(t) = \frac{C(t)}{\int_0^{\infty} C(t) dt} \quad (10)$$

$$t_r = \int_0^{\infty} t E(t) dt \quad (11)$$

$$E(\theta) = t_r E(t) = t_r \frac{C(t)}{\int_0^{\infty} C(t) dt} \quad (12)$$

$$E(\theta) = \left(N \underline{t_i} \right) E(t) = \frac{N(N\theta)^{(N-1)}}{(N-1)!} e^{-N\theta} \quad (13)$$

2.5. Strains and culture medium

Escherichia coli MC4100 was selected as a model organism for testing

the bioreactor's performance (Peters et al., 2003). The corresponding growth curves were obtained by monitoring cell density for 24 h with the aid of the OAS. This strain was cultured in lysogeny broth (Bertani, 1951) at 37 °C, and the growth data was contrasted against previously reported experimental data (Doig et al., 2008). Before culturing, the bioreactors were autoclaved with 170 mL of broth. The culture was started by inoculating a single MC4100 colony.

3. Results and discussion

3.1. Simulation

The velocity distribution profiles for the different airflow values are shown in Fig. 3A for a simulation time of 3.6 s. The results suggest that conditions close to ideal mixing can be achieved for the different operating regimes evaluated. This assumption is partially supported by the absence of large dead zones within the reaction volume. Additionally, it is clear that as the airflow increases, the velocity of the liquid phase decreases without detrimentally impacting fluid circulation inside the downcomer. This can be explained by the limited ability of the gas bubbles to transfer momentum to the liquid phase as their residence time considerably decreases at higher airflow rates. This has been previously observed in hydrodynamics studies of airlift and bubble column bioreactors, and particularly when the liquid holdup becomes dominant (Znad et al., 2004; Rossi et al., 2016).

Fig. 3B shows the shear stress distribution for the liquid inside the bioreactor as a function of the airflow regimes explored and for a simulation time of 3.6 s. In line with the results of the velocity field, the shear stress decreases as the airflow increases. At the lower airflow rates,

the induced movement of the fluid leads to a higher viscous dissipation (Wang and McNeil, 1996; Shamlou et al., 1994; Chisti et al., 1988; Young et al., 1991). This could be potentially disadvantageous due to a reduction of cellular viability; however, the maximum shear stress calculated for the system is about seven orders of magnitude below the required to promote cell disruption (Sauer et al., 1989; Yusaf, 2013; Lange et al., 2001).

Importantly, the liquid coming from the downcomer moves directly towards the air sparger, which “pushes” the bubbles away from the downcomer outlet and, consequently, apart from the downcomer inlet. This phenomenon contributes to keeping them from entering the downcomer, reducing the number of bubbles that could potentially interfere with the optical density measurements.

One of the ideas behind having a clear downcomer was to explore the possibility of monitoring cell density in an automated manner during culture with the aid of an optical sensor. This could be only feasible if, during operation, bubble interference with the light path of the sensing device is reduced to a minimum. Here, we implemented bubble counting in the downcomer (via the particle counter tool) to find the operating conditions under which this approach is viable. According to our results, at a gas flow of 0.8 L/min the number of bubbles remains below 10, for the 3.6 s simulation time, which is acceptable for optical sensing measurements. Putting together the velocity profile, shear stress distribution, and bubble number, it was recommended to operate the bioreactor at low airflow rates.

With respect to the airflow, the simulation indicates that at 0.8 L/min, some noticeable low-velocity zones are present. This is important because the working volume of the bioreactor is 170 mL, thereby resulting in a volume of air under standard conditions per volume of

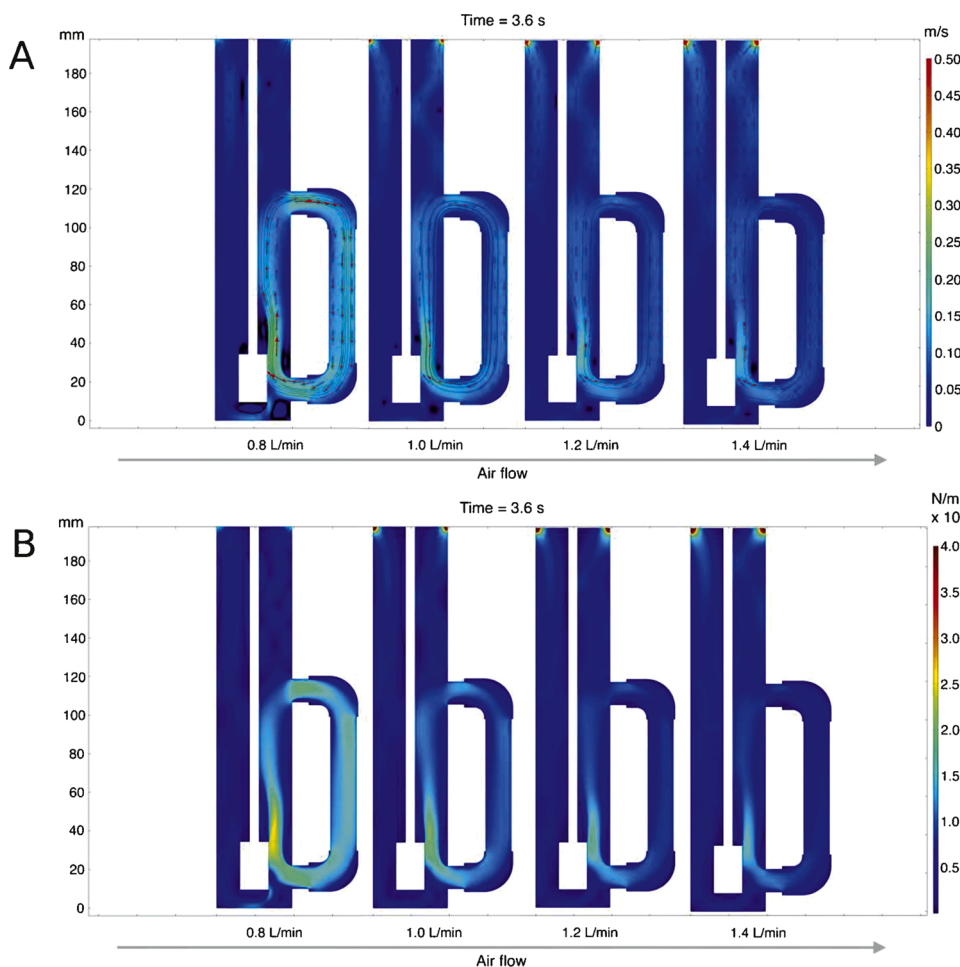


Fig. 3. Analysis of simulation results. A. Velocity field with the streamlines. Conditions close to ideal mixing were achieved for the evaluated airflows, as evidenced by the absence of large dead zones inside the bioreactor. Streamlines (in red) show proper liquid circulation inside the downcomer even at high airflow rates where momentum transfer from the gas to liquid phase decreases. B. Shear stress in the liquid phase (N/m). Shear stress decreases as the airflow is increased due to lower viscous dissipation. The maximum estimated shear stress for the system is seven orders of magnitude below the required for cell disruption. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

liquid per minute (VVM) of 4.7. When compared with similar studies, this is a relatively high value (Shah et al., 1982; Sobhi et al., 2020). As this prototype was intended for operation under mixing conditions close to ideality and the calculated shear stress was not high enough to produce cell damage, a foam control grid was considered to prevent the problems related to foaming and flooding while favoring high turbulence in the bioreactor.

3.2. OAS calibration

Before running the bioreactor, a proof-of-concept OAS prototype was initially assembled and tested, and subsequently compared with readings conducted in a spectrophotometer. In this case, food colorant green #3 was used to prepare 40 calibration solutions with different concentrations. The results showed a linear correlation between the two systems' readings over the entire range of analyzed absorbances (Fig. 4A). These results served as the basis for manufacturing three more high-resolution OAS prototypes.

Calibration of the three high-resolution OAS prototypes proceeded with BG solutions at different concentrations by monitoring absorbance at 420, 630, and 680 nm. The 420 and 630 nm wavelengths were selected because they correspond to absorption peaks of BG. The 630 nm also corresponds to the wavelength of the light emitted by the LED incorporated into the OAS. Moreover, Breidt et al. reported that at this wavelength, it is possible to track optical density changes associated with the cell growth of a wide variety of bacteria (Breidt et al., 1994). Finally, the 680 nm was also included because it is commonly used for monitoring microalgae growth (Vu et al., 2020).

Fig. 4 (three bottom plots) shows a linear correlation between the reading from all the high-resolution OAS prototypes and those of a spectrophotometer. Importantly, although the slopes of the three fitted lines are similar, they are not identical. Also, the intercepts of the fitted lines are entirely different. This could be attributed to slight changes in the positioning and assembly of the device's components. For instance, the relative position of the LED respect to the photoresistor might vary in the order of fractions of a millimeter, which results in subtle changes in absorbance. This indicates that every newly manufactured OAS will have to be calibrated before use. The proposed strategy is to calibrate the OAS using two standard solutions (or suspensions), usually done for a pH-meter calibration. This approach is simple enough to be appealing to most users.

Importantly, all three calibrations resulted in linear tendencies with high Pearson correlation coefficients (R) (Fig. 4). These satisfactory results allowed us to install the OAS on the external loop (downcomer) of the bioreactors.

Besides the relatively simple manufacture and calibration processes of the OAS, these pieces of equipment were designed with the primary goal of making the sensors as inexpensive as possible. This approach aims to help laboratories in emerging countries implement fully equipped bioreactors at affordable costs. This notion has been enabled by the miniaturization of electronics and the availability of low-cost components and microprocessors. As a result, the development of affordable instruments has become increasingly popular, not only for research purposes but also developing and improving modern educational tools and experiences (Lema et al., 2002).

3.3. Tracer test

One of the reasons for replacing shaken flasks for controlled milliliter-scale bioreactors is maintaining more homogeneous conditions for the liquid cultures. Homogeneous mixing is perhaps one of the most important requirements to achieve, and it is well known that shaken flasks generally fail to do so (Klöckner and Büchs, 2012). Furthermore, the most desirable scenario would have not just homogeneous but ideal mixing. Thus, evaluating the mixing regime in the bioreactors is essential to ensure that the proposed design leads to the highest possible

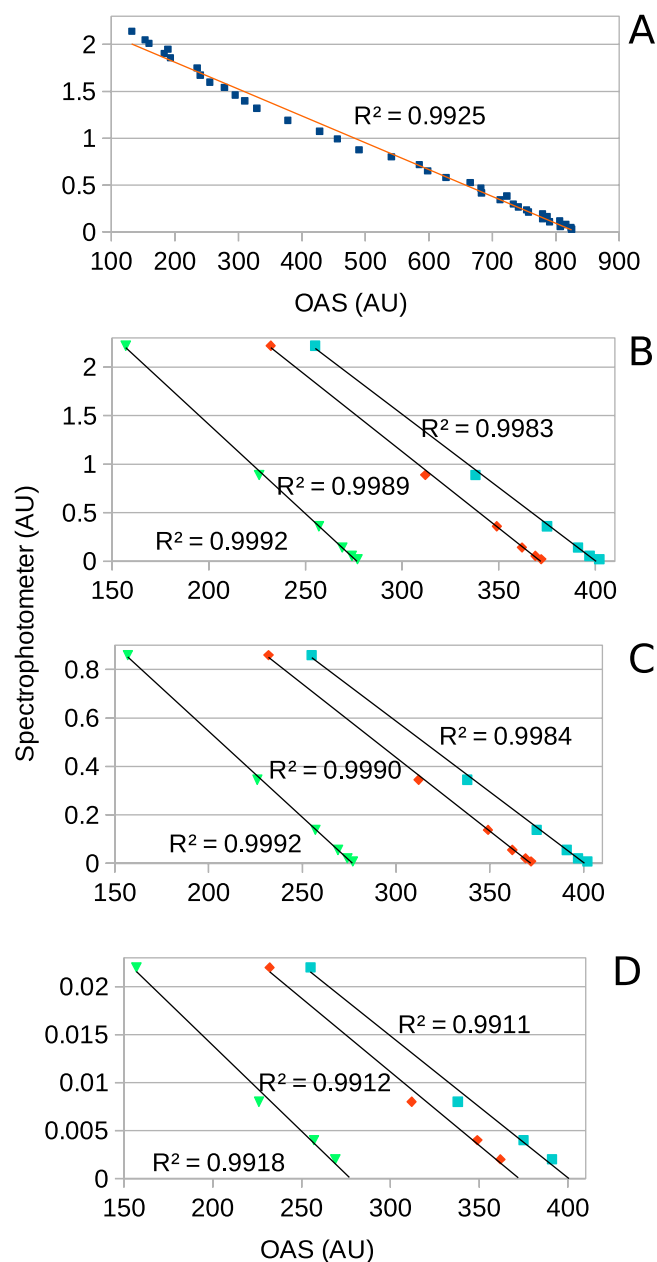


Fig. 4. OAS calibration. A. Linear correlation between the OAS readings and those of a spectrophotometer. This proof-of-concept experiment demonstrated the feasibility of using the OAS to monitor changes in absorbance over an ample range of wavelengths up to 680 nm. The calibration was conducted with aqueous solutions of a commercial food colorant (green #3). B, C, and D. Calibration of three OAS using the dye Bromocresol Green (BG). The linear correlation was maintained for another three high-resolution OAS prototypes. All the calibration experiments were conducted with four aqueous solutions of BG. In this case, the monitored wavelengths were 420 (green inverted triangles), 630 (red diamonds), and 680 nm (blue squares). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

performance. Here, we implemented a tracer test to evaluate mixing, which required us to modify the system for continuous flow operation with a constant feed of water. BC was used as the tracer, and the OAS were modified to fit the bioreactor's exit tubes, so it was possible to measure the BC concentration leaving the bioreactor in real-time. The evaluation was based on a single pulse of the tracer at the time $t = 0$. As both the inlet and the outlet could be located by the top or the bottom of the riser, four different combinations were evaluated. Fig. 5 shows the

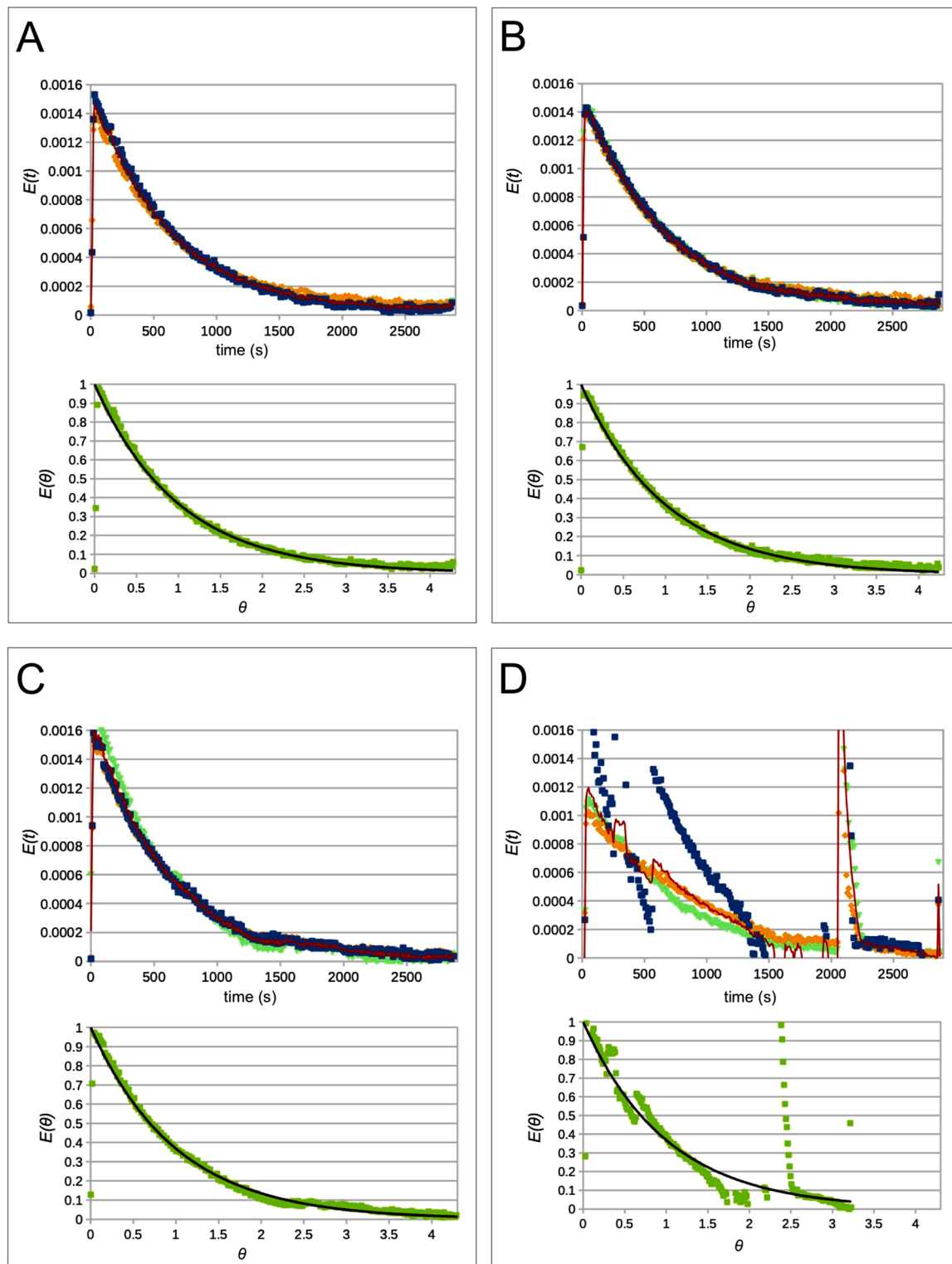


Fig. 5. Evaluation of the HRT distribution in the bioreactor by the tracer test method. Four different configurations were evaluated as two locations for the inlet, and the outlet of the bioreactor were alternated. A. Inlet and Outlet at the Bottom, B. Inlet at the Top and Outlet at the Bottom, C. Inlet and Outlet at the Top, and D. Inlet at the Bottom and Outlet at the Top. The $E(t)$ curves were obtained in triplicate experiments for three parallel bioreactors (yellow, green, and blue). The red line corresponds to the average of the three tests. The $E(\theta)$ curves (green) were calculated from the average $E(t)$ curves. Eq. (9) was fitted to the $E(\theta)$ data with $N = 1$ (black lines). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

resulting triplicate $E(t)$ curves for the four configurations evaluated. By using the averages of the three experiments, a single $E(\theta)$ curve was calculated for each setting and then adjusted to Eq. 9. As it could be inferred from the shape of the $E(\theta)$ curves and the size of the vessels, the

equivalent number of tanks-in-series representing the bioreactors is 1, supporting the initial “close to ideal mixing” assumption and in accordance with the preliminary simulation results.

This experiment corroborated that regardless of the geometrical

considerations, OAS prototypes are quite versatile in tracking subtle changes in absorbance. This suggests that they could be implemented in any reactor since they just have to be connected to the outlet of the system. Also, the fully automated data acquisition from the OAS facilitates the implementation of tracer tests, which is often a tedious task but also permits collecting more accurate and better-defined probability distribution curves $E(t)$. As a result, better goodness of fit to the tanks-in-series model as more precise retention time calculations can be made (Fig. 5).

When analyzing the plotted results, the advantages of operating 3 of the designed bioreactors in parallel became evident. Both, the $E(t)$ and $E(\theta)$ curves resulted almost identical for the three bioreactors, mainly because of the automatization of the optical density data acquisition. Interestingly, there are no sudden shifts in the plots (Hoover et al., 2017; Newell et al., 1998; Olivet et al., 2005) (except for the inlet at the Bottom and Outlet at the Top configuration, Fig. 5D), which facilitates the retention time analysis. The size of the systems and the frequency of the data acquisition is likely to play an essential role in smoothing out the tendencies. The level of detail in the plots can also be seen as an indicator of the confidence level achieved with this relatively simple setup.

For the fourth configuration (Fig. 5D, Inlet at the Bottom and Outlet at the Top), the data tendencies are not as smooth as for the other cases. This was most likely due to the interference of the bubbles exiting the system through the outlet. Similar behavior would have been expected for the third configuration (Inlet at the Top and Outlet at the Top); however, no noticeable interference was detected. These results indicate that when the outlet is at the top of the liquid column, bubbles tend to follow random pathways and could eventually reach the exit tubing, where the OAS was located. Therefore, this configuration is to be avoided for tracer test experiments, and even the regular operation of the bioreactors as the interference of bubbles leads to unreliable optical density data.

Importantly, the ideal mixing regime predicted by CFD simulations was experimentally corroborated for this system. Airlift bioreactors are frequently modeled using the tanks-in-series model (Behin and Ahmadi, 2010), where the equivalent number of tanks in series (N in Eq. (9)) is calculated to describe the real behavior. For most airlift reactors, N is higher than one and commonly higher than 10 (Behin and Ahmadi, 2010; Gavrilescu and Tudose, 1999; Zuo et al., 2006). Consequently, the ideal mixing regime is not necessarily the expected result for this type of system. Here, we found that to model our novel bioreactor, only one tank in series is needed. We hypothesize that a small enough airlift with a relatively large riser compared to the lateral downcomer, would result in the desired ideal mixing regime. Achieving this regime was necessary for the reliability yet simplicity of the automated optical density data acquisition as it would genuinely represent the cell density of the reaction mixture.

3.4. Growth curves

Once the OAS prototypes were successfully tested, and a proof-of-concept tracer test supported close to ideal mixing, the setup was complete to evaluate cell growth under batch operation. Growth curves were then built up for *E. coli* as a model organism in an automated manner via OAS measurements. Prior to the cell growth experiments, the OAS was calibrated with the aid of aqueous suspensions of *E. coli* at different cell densities. In this case, a procedure similar to that of BC calibration was followed (see above for details). Among the factors that can affect the OAS's sensitivity, a resistor in series with the photoresistor (sensor) plays one of the most critical roles. Although this resistor's electrical resistance can be selected arbitrarily, it changes the resolution and permits to adjust the sensitivity of the measurements. In line with this, it is also worth mentioning that although some of the samples analyzed in the spectrophotometer required dilution, the samples analyzed with the aid of the OAS did not; still, a linear correlation between the readings was observed (R^2 values close to 1). Recently,

another optical sensor, directly attached to a bioreactor, was reported to allow OD measurements without the need for sample dilution (Krujatz et al., 2016). This is a key feature enabling the OAS to be used for in-line and real-time OD measurements.

After OAS calibration using *E. coli* cells, three growth curves were obtained under batch conditions (Fig. 6). Importantly, the three curves showed a high level of agreement, which provides further evidence that these bioreactors are ideally mixed. The noninvasiveness of the OAS monitoring system is an attractive feature if dealing with cultures prone to contamination by opportunistic microorganisms. Moreover, the possibility of parallelizing the experiments and data collection without significant major investments provides a convenient route for running proof-of-concept experiments in a reproducible, rapid, and reliable manner.

The data from the three experimental growth curves were averaged and fitted to the Verhulst model (logistic equation) (Shirsat et al., 2014) (continuous blue line in Fig. 6). For the sake of comparison and validation, a dataset previously reported for a similarly sized bioreactor and

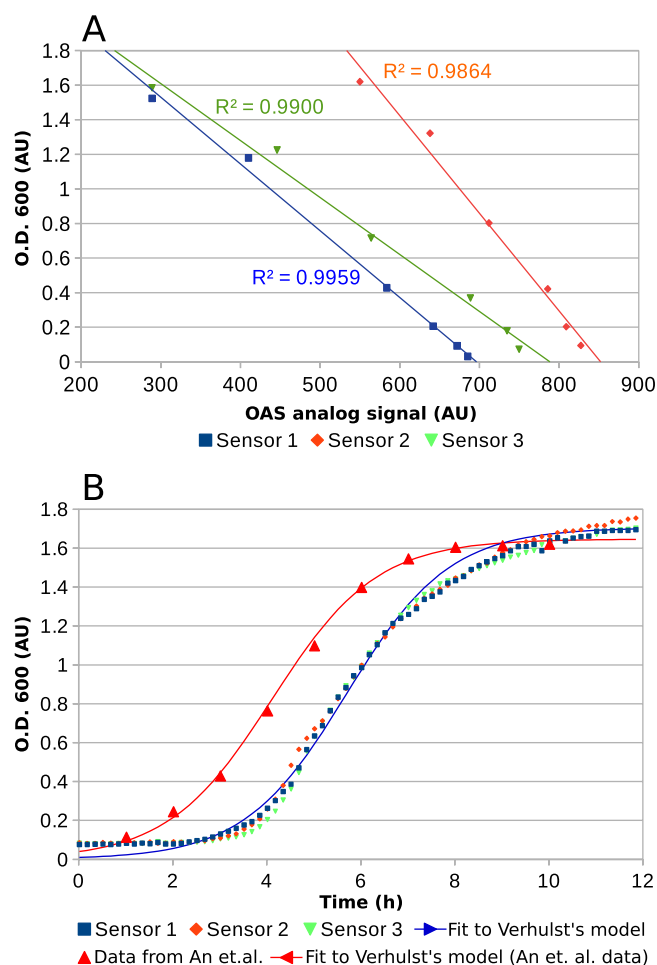


Fig. 6. Evaluation of the bioreactor performance by creating growth curves for *E. coli* in an automated manner. A. A. Correlation between the OAS readings and the spectrophotometer readings. Some curvature can be observed for the upper and lower part of data ranges, but the Pearson correlation coefficients (R^2) were close to 1. Cell growth curves as obtained from the three high-resolution OAS prototypes installed on the downcomer of the bioreactors (green inverted triangles, red diamonds, and blue squares). Verhulst's model was fitted to the experimental data (continuous blue line). The red triangles correspond to previously reported growth data (An et al., 2015). Verhulst's model was also fitted to these previously published data (continuous red line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

the same culture medium (An et al., 2015) was also fitted to the Verhulst model (continuous red line in Fig. 6). A visual inspection of the curves suggests that the exponential phase for both datasets show a similar slope, which can be attributed to a comparable growth rate. Still, the lag phase of our experiments appears longer. This is not the case of the maximum cell concentration, as it seems that both datasets approached the same O.D. level after about eight hours. The recovered parameters from the Verhulst model are presented in Table 1. The lower initial population parameter (P_0) confirms that the more extended lag phase observed in our case led to starting with a much smaller microorganism population. This could be explained by the limited previous exposure of our strain to the lysogeny broth as a culture medium. Also, in line with the described observations above, the saturation population (K), as well as the growth rate constant (r) are similar for both studies.

Importantly, comparing the kinetic constants obtained using our prototype with the constants obtained from another laboratory-scale independent study, permitted the validation of the system under a particular set of conditions beyond the proof-of-concept. The bioreactor was operated with a cell culture fed by a commonly used culture medium. The hydrodynamic conditions here were hence different from those used for the tracer test. Given that the r and K constants were similar for both data sets, this experiment suggests that the mixing conditions were still good enough for collecting robust cell growth data. The experiment also supports the initial premise that this low-cost setup permits the study of cell growth in liquid culture under close to ideal conditions while allowing data acquisition automatically and without affecting the bioreactor vessel's monoseptic conditions. In that order of ideas, our prototype will permit the execution of multiple replicas of a single experimental condition with a relatively low investment (one full setup can be assembled for less than 100 US dollars) and generate reliable data. This facilitates the comparison of different sets of operation conditions, the preliminary optimization of variables, and the collection of ideal condition parameters that can be used as a reference before further scale-up.

4. Conclusions

Scaling-up bioprocesses is generally a tedious and expensive task due to the required time and investment in fully instrumented commercial systems. This has been particularly limiting in settings operating under limited budgets. To address this issue, here we put forward the idea of manufacturing in-house, low-cost, milliliter-scale bioreactors, where proof-of-concept experiments can be run with reduced investment in materials, reagents, and instrumentation. We first started by conceptually designing the system *in silico* by considering the external-loop airlift bioreactor as the most attractive configuration. These studies were based on implementing the conservation of momentum equations in COMSOL Multiphysics (CFD module) for a multiphase system. Also, to track the circulation of bubbles produced by the bioreactor's sparging system, the particle tracing module of COMSOL was implemented. The simulation results indicate that the proposed system provides ideal mixing conditions and low shear stress under certain operating conditions. Moreover, the particle tracing studies allowed us to identify an airflow regime where circulating bubbles passing by the downcomer are kept to a minimum. This information was needed to install, on the downcomer, low-cost prototypes of optical absorbance sensors (OAS) to monitor cell density during an experiment.

The milliliter-scale bioreactor parts were manufactured with the aid of 3D additive manufacturing techniques and subsequently assembled to conduct characterization studies of the bioreactor hydrodynamics. These studies were enabled by installing, calibrating, and successfully evaluating three home-made OAS using tracer substances and a commercial spectrophotometer. The collected OAS data confirmed that, as predicted by the CFD studies, ideal mixing conditions are easily achievable. Further modeling via the N tank-in series approach confirmed that only one tank is needed to capture the bioreactor's real

Table 1

Recovered parameters after fitting to the Verhulst model.

Verhulst model parameters.	This study	(An et al., 2015)
P_0 – initial population	0.0093	0.0400
K – saturation population.	1.7048	1.6461
r – growth rate constant	0.9139	0.9023

behavior, which indicates ideal mixing as well.

Finally, the performance of the bioreactors and the OAS was tested by building a bacterial growth curve. The obtained cell growth results were equivalent to those obtained by others on a similar scale setup in terms of exponential growth rate and maximum achievable density. We are confident that the bioreactors put forward here offer an alternative for conducting proof-of-concept scaling-up experiments at extremely low cost. The possibility of collecting reproducible cell density data at a fast pace is quite appealing to carry out different studies including, response to inhibitors and changes in substrates, presence of consortia, and perhaps most importantly, optimization of productivity.

CRedit authorship contribution statement

Jorge Eduardo Marroquín-Fandiño: Conceptualization, Investigation, Formal analysis. **Carlos Manuel Ramírez-Acosta:** Software, Formal analysis. **Hector Javier Luna-Wandurraga:** Conceptualization, Methodology. **Joaquín Andrés Valderrama-Rincón:** Methodology, Software, Validation. **Juan C. Cruz:** Conceptualization, Methodology, Validation, Writing - original draft, Writing - review & editing. **Luis H. Reyes:** Conceptualization, Methodology, Validation, Writing - original draft, Writing - review & editing. **Juan D. Valderrama-Rincon:** Conceptualization, Methodology, Investigation, Formal analysis, Writing - original draft, Writing - review & editing, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by MINCIENCIAS (Colombia) [grant number 110477757340]. Additional financial support was provided by the Universidad de Los Andes Research Support Grant [FAPA PR.3.2017.4627] to LHR. Also, the authors would like to thank the Departments of Chemical and Biomedical Engineering at Uniandes for providing access to laboratories and computer infrastructure.

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